Sekwang Lee

AI Researcher & Generalist

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Education

Pohang University of Science and Technology (POSTECH) — Feb 2021 – Feb 2027 (Expected)
Undergraduate Student, Computer Science and Engineering (CSE) — *Pohang, South Korea*

- GPA: 3.92/4.3 (**Summa Cum Laude**)

KTH Royal Institute of Technology — Jan 2026 – Jun 2026
Exchange Student, Electrical Engineering and Computer Science (EECS) — *Stockholm, Sweden*

Publications

[1] **Sekwang Lee** and Ilwoo Lyu. ***AdaPhase: Navigating the Manifold with Adaptive Frequency Modulation for Faithful Diffusion Stylization***, under review.

Research Experience

Research Intern, Medical Information Processing Lab - CG/CV Team — Jan 2025 – Feb 2025 • Sep 2025 – Dec 2025
Department of Computer Science and Engineering, POSTECH — *Pohang, South Korea*

- Advisor: Prof. Ilwoo Lyu
- Reviewed literature on **spherical deep learning** • Conducted research project: ***AdaPhase*** [1]

Research Intern, Computer Vision Lab — Aug 2023 – Nov 2023
Department of Computer Science and Engineering, POSTECH — *Pohang, South Korea*

- Advisor: Prof. Suha Kwak
- Reviewed literature on **Fourier-based domain generalization** • Formulated research topic: ***AdaIN-based Generalization***

Projects

Distillation Through Attention of SwinDETR — Jul 2024 – May 2025
Undergraduate Research Program • Deep Learning (Graduate Course), POSTECH

- Worked as team leader (A+ in the course)
- Designed ***SwinDETR*** for small-object detection • Introduced **distillation queries**

Autonomous Drone System with MIMO Signal Processing and Multimodal AI — Mar 2024 – Dec 2024
I-Corps Program, Ministry of Science and ICT

- Worked as AI Developer
- Designed ***FIFO-Transfuser*** **(image-radar multimodal)** for foggy conditions • Conducted field research at two fire stations

Work Experience

Student Advisor — Feb 2025 – Dec 2025
Mueunjae School of Undergraduate Studies, POSTECH — *Pohang, South Korea*

- Conducted **66 counseling sessions** for freshmen • TA for freshman courses: Future Planning for College Life I & II

Public Relations (PR) Coordinator — Sep 2024 – Dec 2025
Institute of Communication and Public Discourse, POSTECH — *Pohang, South Korea*

- Led promotional content creation • Designed program posters (**founding member, invited by professor**)

Software Engineering Intern, AX Technology Team — Jun 2025 – Aug 2025
LG Electronics — *Seoul, South Korea*

- Designed and implemented ***OneClick***: a website task automation service (**founding member**)

Tech-Review Manager — Jul 2024 – Jul 2025
Start-up Support Team, POSTECH — *Pohang, South Korea*

- Hosted technology seminars with startup CEOs • Organized a technology-based ideathon

Military Police (Military Service) Jan 2022 – Jul 2023
Republic of Korea Army • Ministry of National Defense *Seoul, South Korea*
- Worked as squad leader (5 months)

Awards & Honors

Best Research Project Award (Grand Prize) Dec 2025
Departent of Computer Science and Engineering, POSTECH
- Conducted research project: ***AdaPhase*** [1]

Next-generation Engineer Award Aug 2025
Institute for Promotion of Engineering and Science of Korea (IPESK)

Presidential Science Scholarship Jun 2025
Ministry of Science and ICT

Global Leadership Program Sep 2024
Department of Computer Science and Engineering, POSTECH
- Merit-based scholarship for the top 8 CSE students

Book Review Essay Contest (Grand Prize) Aug 2024
Institute of Communication and Public Discourse, POSTECH
- Wrote on the social embeddedness of science and scientists' responsibility for scientific uncertainty

IT & Game Scholarship Jul 2024
Sangah Foundation

POSTECH Programming Contest (3rd Prize) May 2024
Department of Computer Science and Engineering, POSTECH

Young Engineers Honor Society (YEHS) Jan 2024
National Academy of Engineering of Korea (NAEK)

Special Experience

ARPA-E Energy Innovation Summit Mar 2025
Advanced Research Projects Agency-Energy (ARPA-E) *Washington, D.C., USA*
- Attended as a POSTECH student

Korea Cross Country Cycling Road Jul 2024
Ministry of Environment • Ministry of the Interior and Safety *South Korea*
- Cycled 633 km over 6 days from Busan to Incheon (completed the cross country cycling certification)

Book Published: *I Can Live a Wonderful Day Today* (『나는 오늘도 멋진 날을 살 수 있다』) Jun 2023 – Feb 2024
Self Project
- Authored a 190-page collection of stories and reflections during military service

CES 2024 Jan 2024
Consumer Technology Association (CTA) *Las Vegas, USA*
- Attended as a POSTECH student